

MA 301
Test 3
Spring - 2013

Problem	points	out of
1		10
2		10
3		10
Score		30

Name: _____

HR: _____

Signature: _____

Formula Sheet

The Fourier series of a function, $f(x)$, with period, $2L$, is given by

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$$

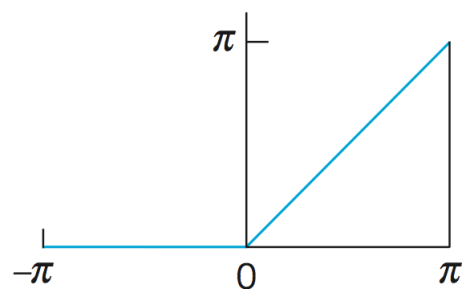
where

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$$

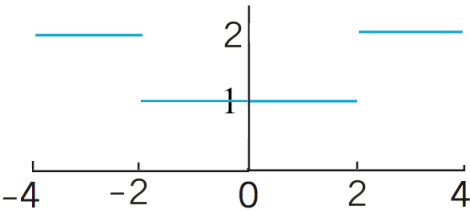
$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

Problem 1: Determine the Fourier Series for the function depicted in the graph below. Assume it periodically extends along the x -axis.



Problem 2: Determine the Fourier Series for the function depicted in the graph below. Assume it periodically extends along the x -axis.

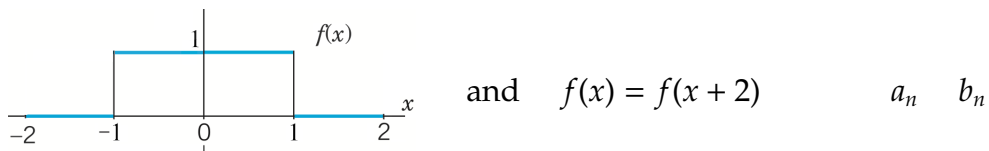
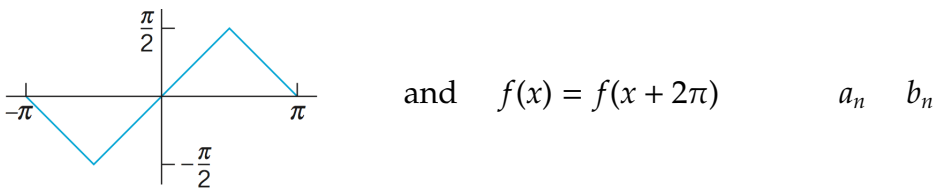
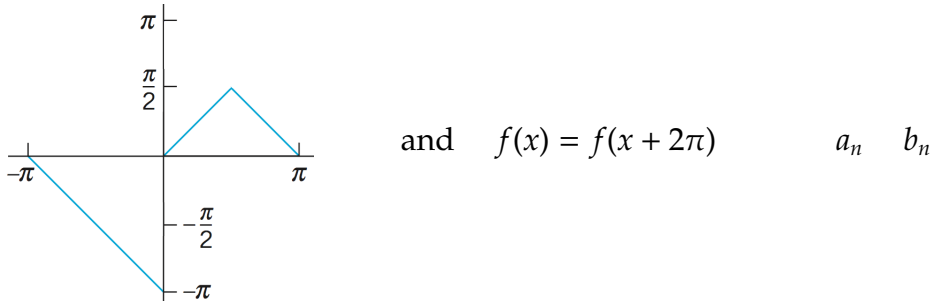
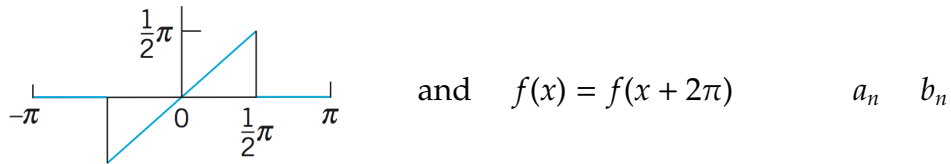


Problem 3: For each function $f(x)$ below, circle the Fourier Coefficients that must be zero.
 Let $g(x)$ be an even function of period 1 and $h(x)$ an odd function with period 1.

$$f(x) = \begin{cases} 0 & -2 \leq x \leq 0 \\ 1 & 0 \leq x \leq 1 \\ 0 & 1 \leq x \leq 2 \end{cases} \quad \text{and} \quad f(x) = f(x+4) \quad a_n \quad b_n$$

$$f(x) = \begin{cases} -2 & -\frac{1}{2} \leq x \leq 0 \\ 2 & 0 \leq x \leq \frac{1}{2} \end{cases} \quad \text{and} \quad f(x) = f(x+1) \quad a_n \quad b_n$$

$$f(x) = |x| \quad \text{and} \quad f(x) = f(x+1) \quad a_n \quad b_n$$



$$f(x) = 2g(x) \quad \text{and} \quad f(x) = f(x+1) \quad a_n \quad b_n$$

$$f(x) = g(x) \cdot h(x) \quad \text{and} \quad f(x) = f(x+1) \quad a_n \quad b_n$$

$$f(x) = \cos(h(x)) \quad \text{and} \quad f(x) = f(x+1) \quad a_n \quad b_n$$